

Part C — Comparative Analysis

Image Generation using (β -)VAE, VQ-VAE and WGAN

Group Number: 123

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Image generation on CIFAR-10 (32×32 RGB, pixel values normalised to [0,1]). This report compares the three generative families trained in Parts A and B on reconstruction error, sampling quality and speed. Every number below was produced by the Part A and Part B notebooks under one shared evaluation protocol, and is read directly from results/metrics_partA.json and results/metrics_partB.json.

Evaluation protocol

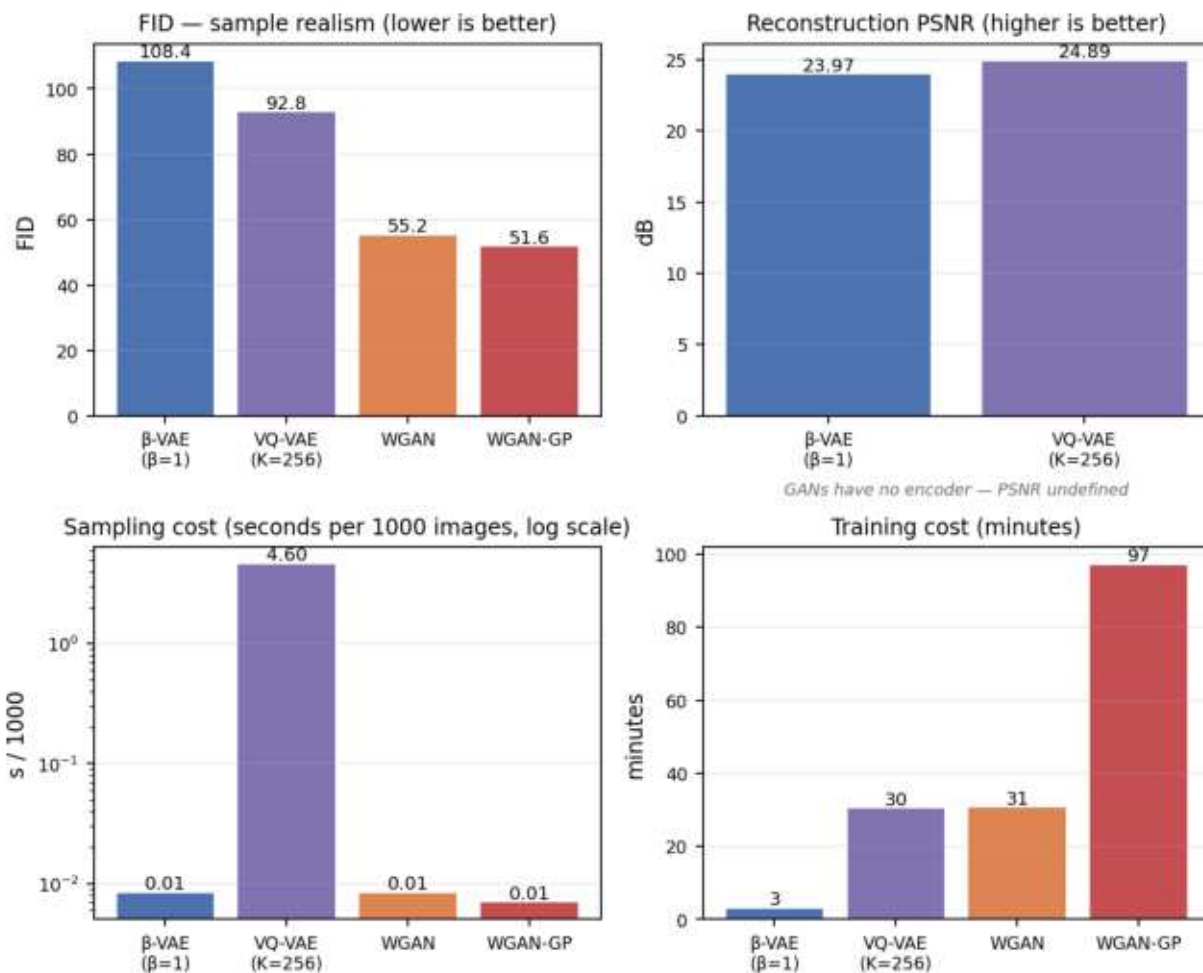
- FID — torchmetrics Fréchet Inception Distance on 2048-d InceptionV3 pool features. Reals are the 10,000 CIFAR-10 test images; fakes are 10,000 generated images. The identical real-side statistics are reused for every model, so the numbers are directly comparable.
- PSNR — mean per-image peak signal-to-noise ratio over the 10,000 test images with data_range = 1.0. It measures reconstruction, so it is only defined for the two autoencoder families; a GAN has no encoder and cannot reconstruct a given image.
- Speed — wall-clock training time on one RTX 4060 Ti, and sampling time per 1000 generated images.

Headline result

On sample realism the ordering is unambiguous: WGAN-GP (gradient penalty) attains the best FID (51.62), and both Wasserstein GANs beat both likelihood-based models. On reconstruction the ordering reverses — VQ-VAE + PixelCNN (K=256) gives the best PSNR (24.89 dB), and the GANs cannot compete because they cannot reconstruct at all. On speed the spread is the widest of the three axes: sampling ranges from 0.01 s to 4.60 s per 1000 images, a factor of about 662×. No single model wins on all three axes, which is the substance of the comparison rather than an inconvenience.

Master comparison

model	FID ↓	PSNR (dB) ↑	s / 1000 samples	train (min)	params
β -VAE ($\beta = 1$)	108.36	23.97	0.01	3	2,896,516
VQ-VAE + PixelCNN (K=256)	92.84	24.89	4.60	30	3,816,515
WGAN (weight clipping)	55.16	—	0.01	31	1,183,616
WGAN-GP (gradient penalty)	51.62	—	0.01	97	1,183,616



The four axes disagree with each other, and that disagreement is the result. FID and PSNR rank the models in almost opposite orders, and sampling cost is dominated by a single architectural choice — whether generation is one forward pass or a sequence of them — rather than by model size. Each axis is unpacked in the sections that follow.

100 samples — β -VAE ($\beta = 1$)

FID 108.36 · reconstruction PSNR 23.97 dB · 0.01 s per 1000 samples

β -VAE prior samples, $\beta = 1$ (FID 108.4)



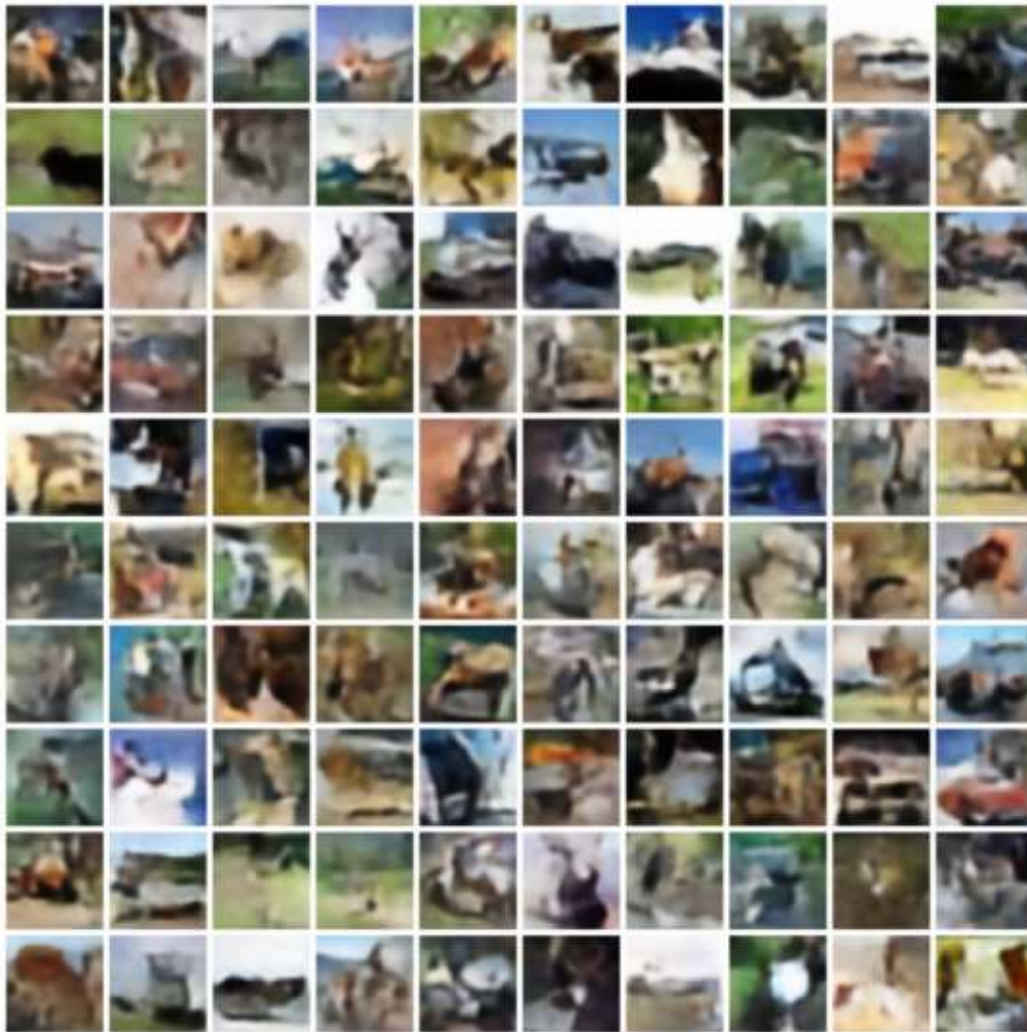
Perceptual quality

The β -VAE samples are globally coherent but perceptually soft: recognisable figure-ground composition, plausible colour statistics and a clear horizon or central-object layout, but almost no high-frequency detail. This is a direct consequence of the objective rather than a training shortfall. A Gaussian likelihood makes the decoder predict $E[x|z]$; when several plausible textures share a latent code, the loss-minimising output is their average, and averaging textures produces blur. The prior samples are noticeably worse than the reconstructions from the same model, which localises the remaining gap in the prior/posterior mismatch rather than in decoder capacity.

100 samples — VQ-VAE + PixelCNN (K=256)

FID 92.84 · reconstruction PSNR 24.89 dB · 4.60 s per 1000 samples

VQ-VAE + PixelCNN samples, K = 256 (FID 92.8)



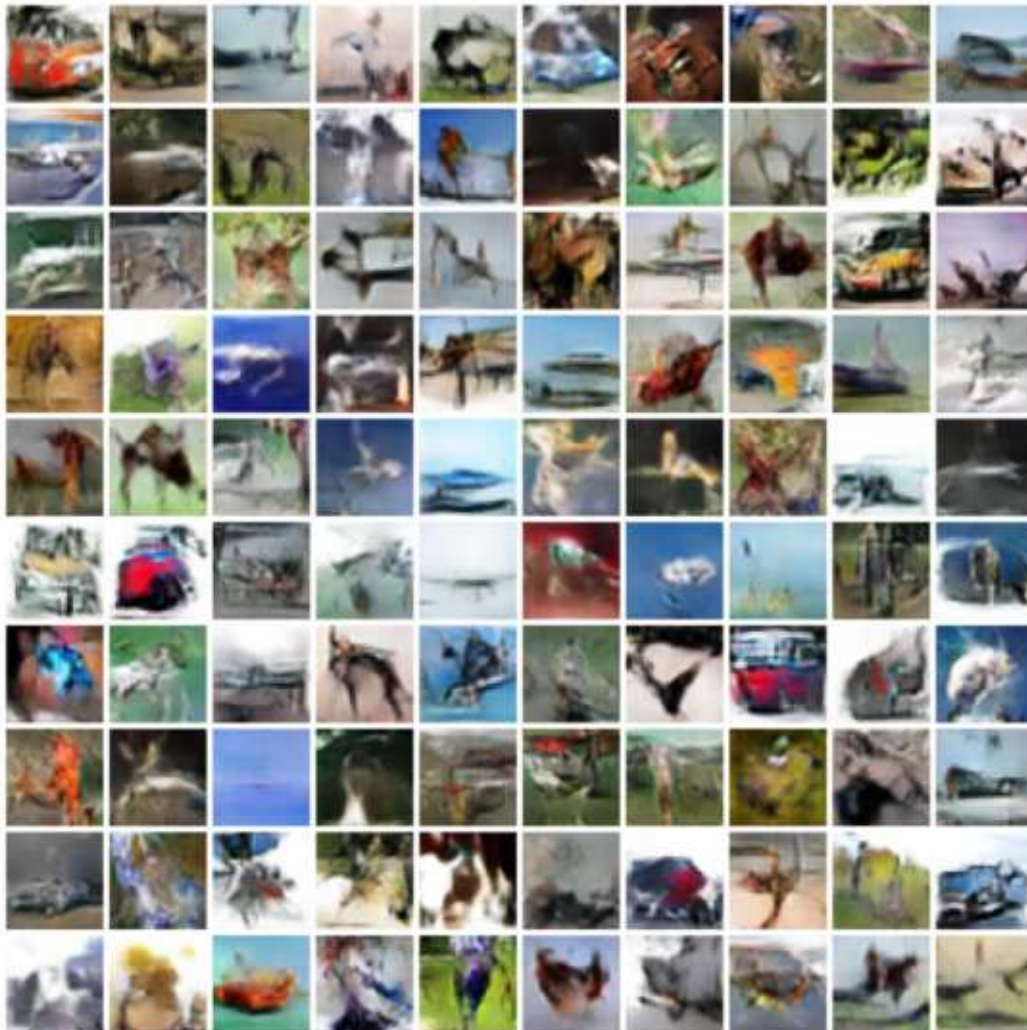
Perceptual quality

The VQ-VAE samples are sharper and more textured than the VAE's — the discrete bottleneck with a straight-through estimator pays no KL tax, so the decoder is free to emit high-frequency detail — but they are less globally coherent. Objects begin to form and then fail to resolve into a single consistent scene. The reason is that these samples are only as good as the PixelCNN prior over the 8×8 code grid: any error in the autoregressive model compounds across the 64 sequential decisions, and a locally plausible code sequence can still be globally inconsistent. Note the asymmetry this creates — VQ-VAE has by far the best reconstruction quality of any model here, yet that fidelity does not transfer to its samples, because reconstruction and sampling exercise different components.

100 samples — WGAN (weight clipping)

FID 55.16 · 0.01 s per 1000 samples

WGAN — 100 random samples (FID 55.2)



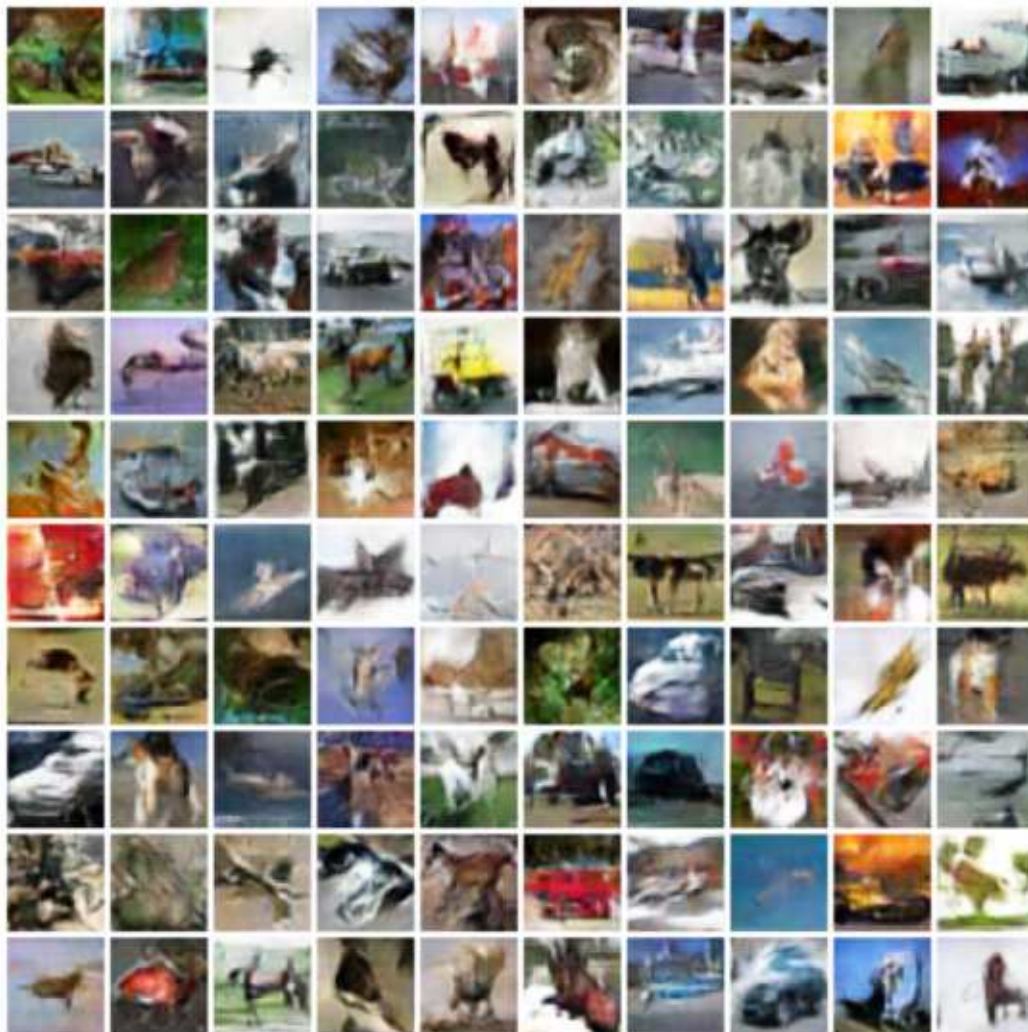
Perceptual quality

The clipped WGAN produces markedly sharper, higher-contrast images than either autoencoder, with convincing local texture. There is no per-pixel reconstruction term anywhere in the objective, so nothing pushes the generator toward a conditional mean. The weakness is consistency: object boundaries are frequently incoherent and some samples degenerate into texture without structure. Weight clipping is a blunt way to enforce the Lipschitz constraint — it caps every weight independently, biasing the critic toward much simpler functions than the constraint actually requires, which weakens the gradient signal reaching the generator.

100 samples — WGAN-GP (gradient penalty)

FID 51.62 · 0.01 s per 1000 samples

WGAN-GP — 100 random samples (FID 51.6)



Perceptual quality

WGAN-GP gives the most realistic samples in the study. Compared with the clipped WGAN the improvement is in global structure rather than local sharpness: more samples resolve into a single coherent object against a consistent background, and far fewer collapse into structureless texture. Penalising the gradient norm only along the line between the real and generated distributions leaves the critic free to use its full capacity, so it supplies a more informative signal. Sample diversity also holds up — the latent interpolations in Part B move smoothly between distinct images rather than cross-fading, which argues against mode collapse.

Reconstruction error

Only the two autoencoder families can reconstruct a given image, so this axis separates them from the GANs entirely. The rows below are the originals followed by each model's reconstruction of them.

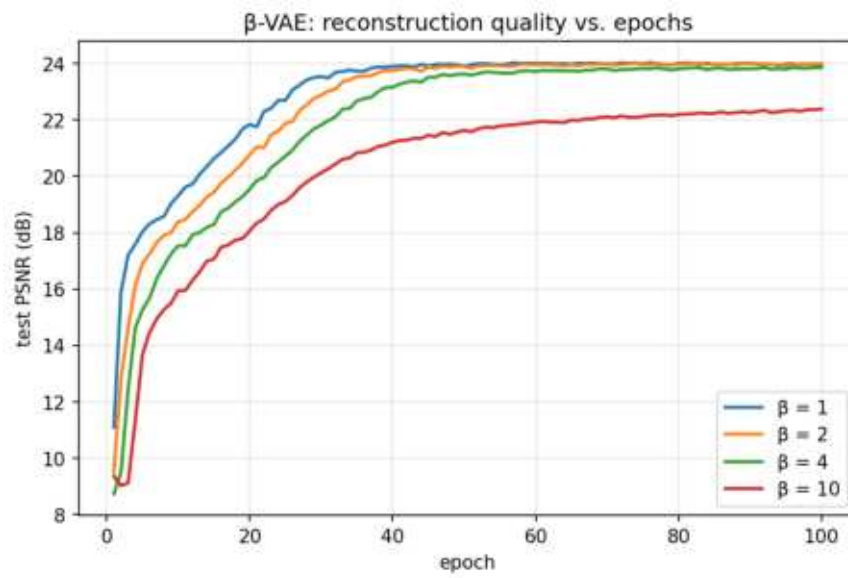


β -VAE: original, then reconstructions at $\beta = 1, 2, 4, 10$



VQ-VAE: original, then reconstructions at $K = 512, 256, 128$

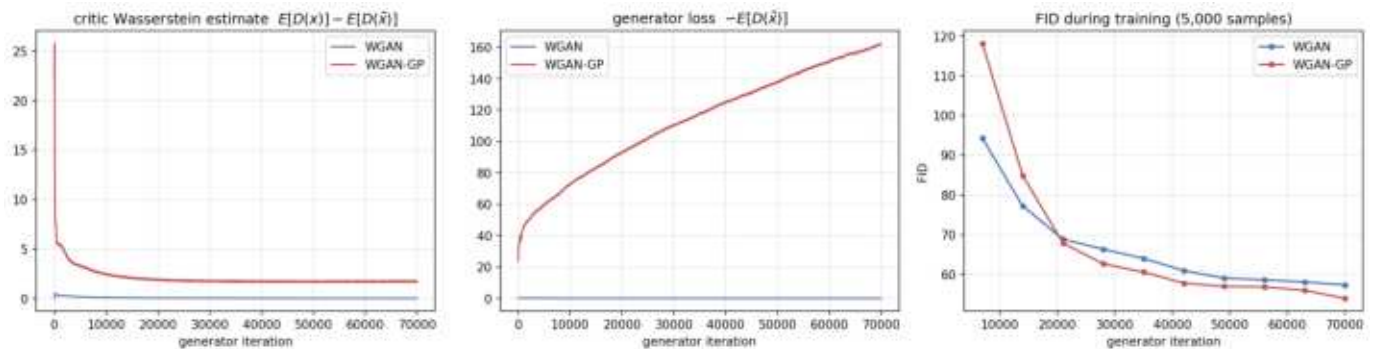
The VQ-VAE reconstructs substantially better than any β -VAE. Both compress to a bottleneck, but the β -VAE additionally pays a KL penalty that pulls the posterior toward the prior and destroys information the decoder needs, whereas the VQ-VAE's discrete code is trained only to be reconstructable. Within the β -VAE family the KL term falls monotonically as β rises ($287 \rightarrow 233 \rightarrow 176 \rightarrow 89$ nats), which is the information bottleneck tightening exactly as the objective intends — but PSNR does not follow it monotonically. PSNR is flat to within 0.15 dB across $\beta = 1, 2, 4$ and only breaks at $\beta = 10$. With a calibrated decoder the 128-dimensional latent has capacity to spare at this resolution, so moderate β removes information the decoder was not relying on; the bottleneck becomes visible in pixel space only once β is large enough to bind.



Test PSNR against training epoch for each β

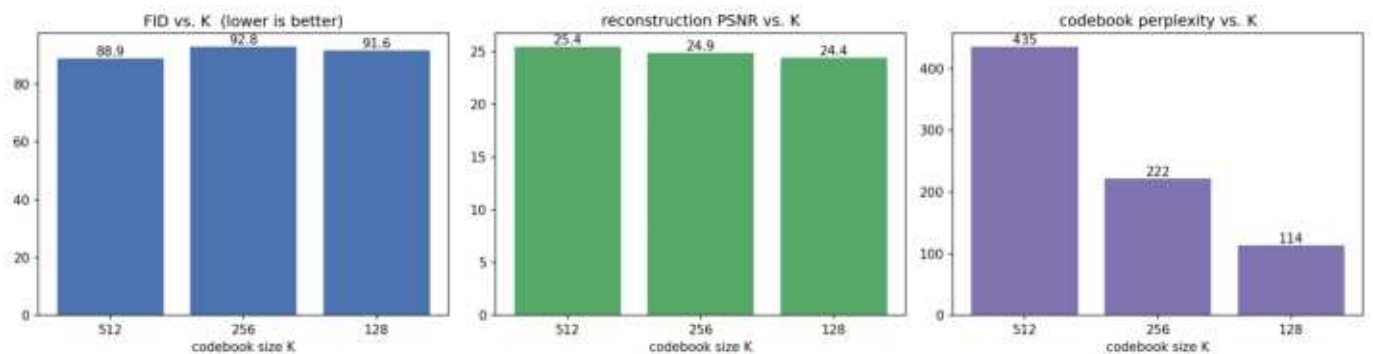
Sampling quality

FID orders the four models as: WGAN-GP (gradient penalty) 51.62, WGAN (weight clipping) 55.16, VQ-VAE + PixelCNN (K=256) 92.84, β -VAE ($\beta = 1$) 108.36. The gap between the adversarial and likelihood-based models is the clearest signal in the study, and it has a specific cause. FID is computed in InceptionV3 feature space, which is sensitive to exactly the high-frequency texture statistics that a per-pixel Gaussian likelihood averages away. A critic trained to separate real from generated images penalises that implausibility directly; a reconstruction loss actively rewards it. This is why FID and PSNR disagree so sharply here, and why reporting either alone would be misleading.



WGAN vs WGAN-GP: critic Wasserstein estimate, generator loss, and FID during training

The two Wasserstein estimates are not on a common scale — weight clipping caps the critic's Lipschitz constant at an unknown value well below 1, so its estimate is a scaled-down proxy. What is comparable is the FID trace measured on the same axis, which shows WGAN-GP both converging faster and reaching a better optimum.



VQ-VAE across codebook size K: FID, reconstruction PSNR, and codebook perplexity

Codebook quality across K

codebook	FID ↓	PSNR ↑	perplexity	codes used	entropy (bits)	PixelCNN NLL
K = 512	88.89	25.45	435	512/512	8.77	7.42
K = 256	92.84	24.89	222	256/256	7.79	6.67
K = 128	91.59	24.43	114	128/128	6.83	5.81

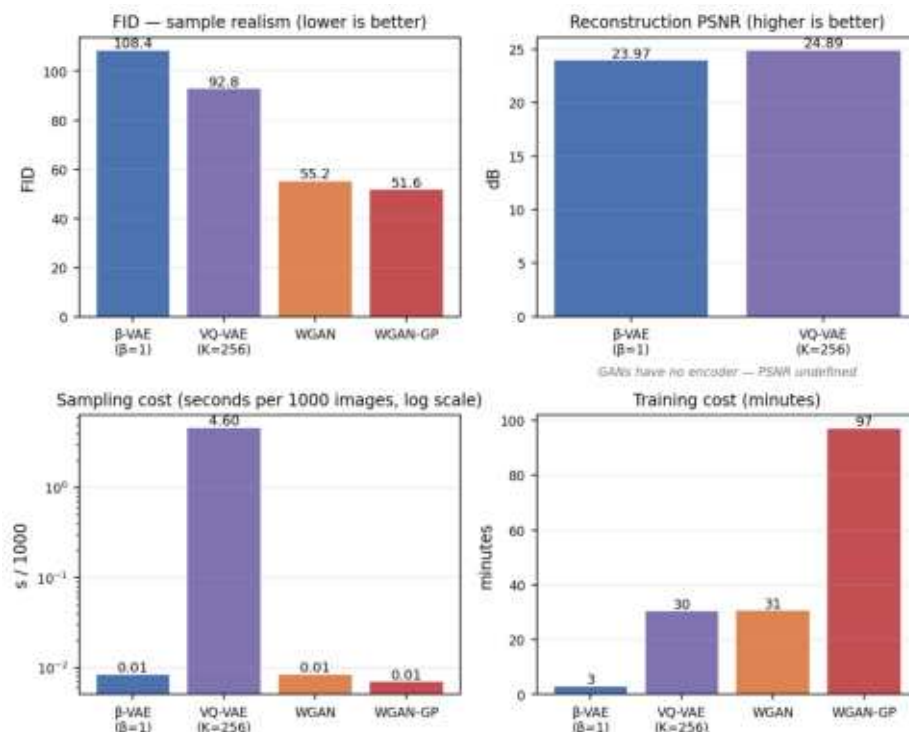
Utilisation is effectively complete at every K , which is not automatic: an EMA codebook initialised from Gaussian noise collapses to a couple of live codes within one epoch, because a code that is never selected is never updated and the EMA divisor then pushes it further from the data. Seeding the codebook from real encoder outputs and restarting dead codes onto live encoder outputs removes the failure entirely (see the quantiser in Part A). Larger K buys better reconstruction, but it also makes the prior's job harder — there are more symbols to model over the same 64 positions — so sample FID does not improve as monotonically as reconstruction PSNR does.

Speed

model	training (min)	s / 1000 samples	images / second
β -VAE ($\beta = 1$)	3	0.01	119,219
VQ-VAE + PixelCNN (K=256)	30	4.60	217
WGAN (weight clipping)	31	0.01	119,351
WGAN-GP (gradient penalty)	97	0.01	143,770

Sampling cost separates the models by roughly 662 $\times$ and the reason is architectural, not a matter of tuning. The β -VAE and both GANs are single-pass samplers: draw z from the prior and run one forward pass, so a whole batch is produced at once. The VQ-VAE is not — generating one image requires 64 sequential PixelCNN evaluations, one per position of the 8 $\times$ 8 code grid, and each must wait for the previous one. That sequential dependency cannot be parallelised away within an image, which makes VQ-VAE the slowest sampler here by a wide margin despite having the smallest decoder.

Training cost tells a different story. The GANs are the most expensive to train, since each generator update is preceded by five critic updates, and WGAN-GP is costlier still because the gradient penalty needs a second-order derivative through the critic. The autoencoders train comparatively cheaply, but the VQ-VAE's total includes training the PixelCNN prior afterwards, which is a second full training run.



Conclusions

What each model is actually good at

- β -VAE ($\beta = 1$) — a stable, cheap model with an encoder, meaningful reconstruction and single-pass sampling. Its samples are the blurriest of the four (108.36 FID), and that blur is inherent to maximising a Gaussian likelihood, not a training artefact. Raising β costs latent information monotonically, but only degrades FID and PSNR once β reaches 10.
- VQ-VAE + PixelCNN ($K = 256$) — the best reconstruction in the study (24.89 dB) and a genuinely useful discrete representation, at the cost of the slowest sampling by two orders of magnitude and a two-stage training pipeline. Its sample quality is limited by the prior, not the autoencoder.
- WGAN — sharp samples and a stable objective, but weight clipping wastes critic capacity and it is beaten by WGAN-GP on FID (55.16 vs 51.62).
- WGAN-GP — the best sample realism here (51.62 FID) and fast single-pass sampling, but no encoder, no likelihood, and the highest training cost.

The general lesson

The three families are not competing on one axis; they are making different trades against the same budget. Likelihood-based models optimise a per-pixel reconstruction term, which gives them encoders, stable training and interpretable objectives — and simultaneously guarantees blur, because averaging over plausible outputs is what minimises that loss. Adversarial models optimise a learned distributional distance, which buys realism precisely where the pixel loss fails, and pays for it by discarding the encoder and the likelihood. The VQ-VAE is the informative middle case: it removes the KL tax and recovers excellent reconstruction, which shows that the VAE's blur comes from its objective rather than from discretisation or limited capacity — but it then inherits a new bottleneck in the prior, and pays for it in sampling speed.

Practically, the choice follows from what is needed. For compression, retrieval or any task requiring an encoder, the VQ-VAE is the strongest option. For generating large volumes of realistic images, WGAN-GP dominates on both quality and throughput. The β -VAE remains the right default when training stability and a tractable objective matter more than perceptual quality.